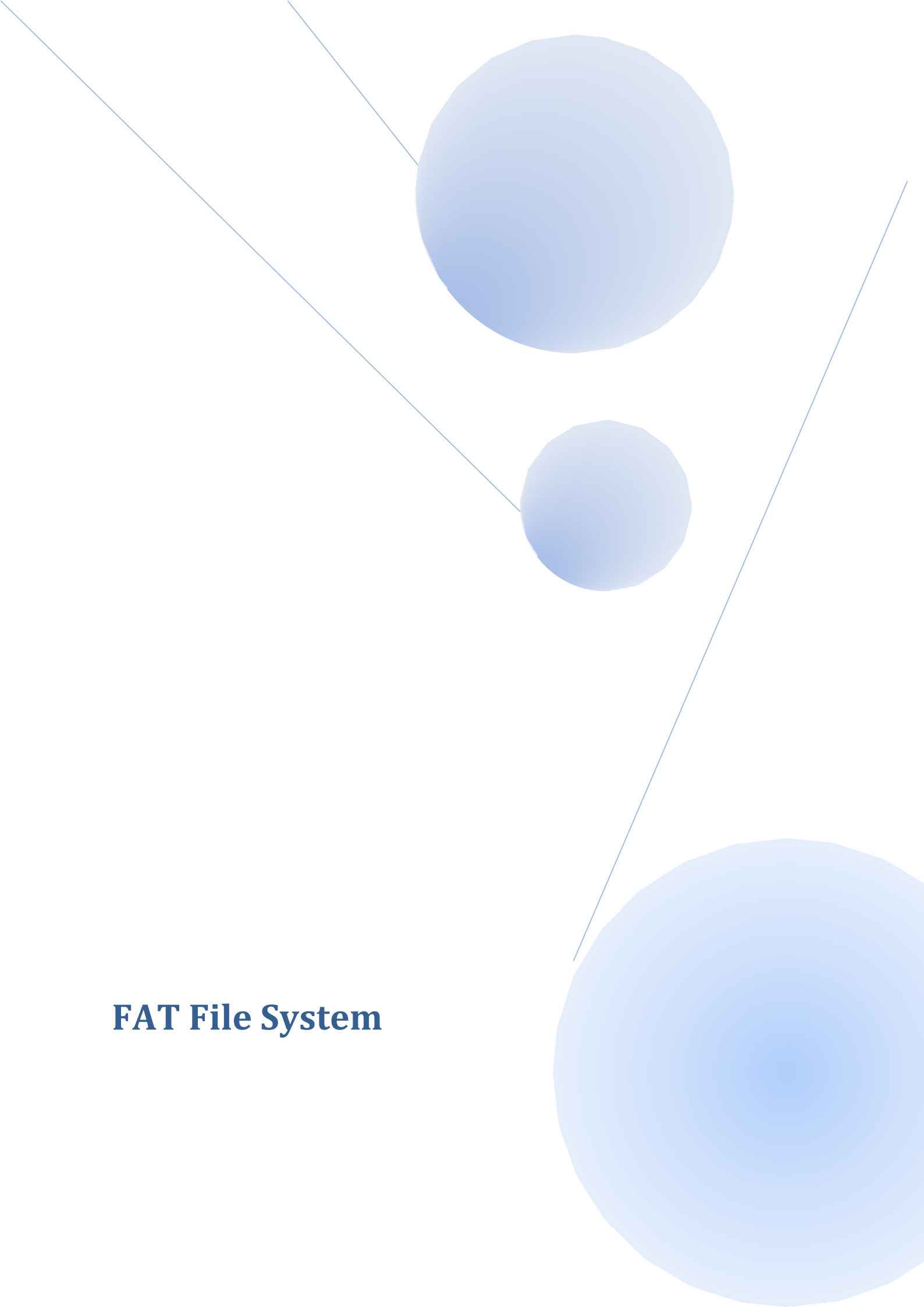


The background features three blue circles of varying sizes and three thin blue lines. One line connects the top-left corner to the top circle. Another line connects the top-right corner to the middle circle. A third line connects the bottom-right corner to the bottom circle. The circles are positioned in the upper and lower right areas of the frame.

FAT File System



Overview



File System

FAT File System

Write Process

Read Process

What happens when a file is deleted?

How to recover the deleted file?

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This session covers the fundamentals of file systems and specifically on FAT file systems.

1. File System
2. FAT File System
3. Write Process
4. Read Process
5. What happens when a file is deleted?
6. How to recover the deleted file?



What is a File System?



- **A File System consist of 2 words – File and System.**
 - File is a bunch of Data put some where.
 - Instead of just storing data anywhere, we can generalize and construct a system which will virtualizes for us a nice and ordered structure in which we could arrange our data.
 - The purpose of file system is to make it easy for us to update and maintain our data.
 - It is the task of file system management code to store book keeping information which will serve the OS for showing us the nice and ordered structure.

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In a computer system, the processes and information resides in the primary memory while processing. This memory can only store temporarily. So we use secondary storage device to store the information permanently. Any information stored in an organized manner is a file. For example, list of all employees in an organization, a web page saved in the system, a picture, an audio file etc. Files in a system are organized and managed by a file system. File management is one of the functionalities of the operating system.

A File System consists of 2 words – File and System. We all know File is a bunch of Data put somewhere. Instead of just storing data anywhere, we can generalize and construct a system which will virtualizes for us a nice and ordered structure in which we could arrange our data.

The purpose of file system is to make it easy for us to update and maintain our data. It is the task of file system management code to store book keeping information which will serve the OS for showing us the nice and ordered structure.



File Systems



FAT12/16/32
NTFS
EXT2FS, EXT3FS
UFS
CDFS

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File system is the set of data structures (like table) and software to manage all the files in the system. Based on the organization of files, we have different file systems like FAT12, FAT16, FAT32, NTFS, EXT2FS, EXT3FS, UFS, CDFS etc.



Lifecycle of Disk Drive



Blank media
Low level format
 Performed at factory
Partition
High level file system format
Installation of Operating system
System operations

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Life cycle of storage disk is first low level formatting is done a blank storage media performed at the factory itself. High level formatting is carried over after the partition is made. File system is created based on the operating system used and the File system mentioned.

After the file system is created the operating system will be installed and do the system operations.



Lifecycle of Disk Drive



Blank media
Low level format
 Performed at factory
Partition
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Installation of Operating system
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After the file system is created the operating system will be installed and do the system operations.



Partitions



- Low-level formatting is done at the factory. It marks off the disk into sectors and cylinders, and defines their placement on the disk.
- **FDISK** is used to partition a hard drive (high-level).



Partitions...



- When the hard drive is divided into more than one partition, the first partition is referred to as the primary partition and the second is called the extended partition.
- The primary partition is usually the active partition, or the partition Windows refers to during the boot-up process.
- A hard disk can have up to 4 separate primary partitions, or 3 primary and 1 extended

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Partitions...



- Fat16 can only address a maximum of 2 gigabytes (GB) of hard disk space.
- FAT32 increases the limit to 2 terabytes (TB) of hard disk space.
- A basic drive must contain atleast one and not more than four primary partitions
- Primary partition is the active partition
 - contains files to start the Operating System

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Fat16 can only address a maximum of 2 gigabytes (GB) of hard disk space.

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A basic drive must contain at least one and not more than four primary partitions Primary partition is the active partition

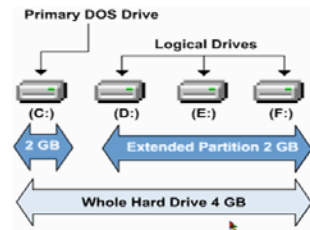
contains files to start the Operating System



Partitions...



- Extended can be subdivided into a maximum of 23 sections called logical drives.
- In Windows, in addition to the drive letter, each drive can be assigned a name called a drive label. The drive label can be up to 11 characters long.



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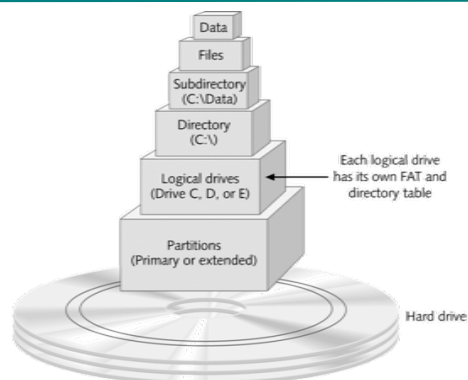
10

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Partitions...



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Cluster



- Group of Sectors
- Cluster size is of 512/1024/2048/4096....
- Large clusters waste disk space because only a single file can live in a cluster
- Small clusters make it hard to allocate clusters to files contiguously and lead to large FAT.

When a partition is formatted, the sectors inside the partition will be grouped in to clusters or blocks. Cluster is group of sectors. Cluster size can be specified at the time of formatting the disk. Cluster size is of 512/1024/2048/4096.... (1/2/4/8... sectors/cluster).

Since cluster is the basic allocation unit, a small file residing on a large cluster wastes disk space. On the other hand, if cluster size is too small, cluster allocation and file access will consume more time.



Allocation Method



An allocation method refers to how disk blocks are allocated for files:

Contiguous allocation

Linked allocation

Indexed allocation

The file system treats storage space as a collection of clusters.

When a user creates a file, the file system assigns the required number of clusters. This process is known as disk allocation. Clusters are the basic allocation units. The different allocation methods are

Contiguous allocation

Linked allocation

Indexed allocation.



Contiguous Allocation



Each file occupies a set of contiguous blocks on the disk.

Simple – only starting location (block #) and length (number of blocks) are required.

Wasteful of space (dynamic storage-allocation problem).

Files cannot grow.

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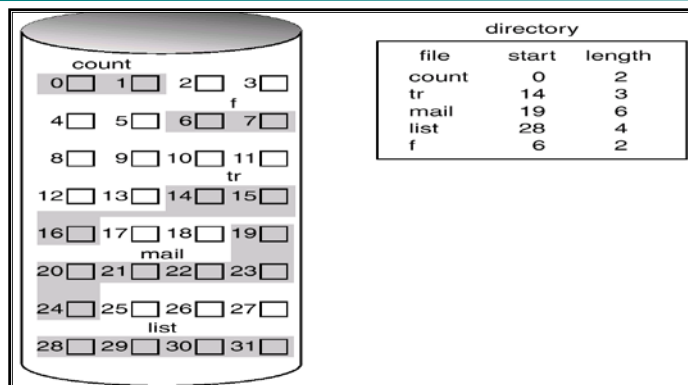
14

In contiguous allocation, contiguous or adjacent blocks will be allocated to files. For example, if a file requires 25 blocks then disk will be searched for 25 free contiguous blocks and the file will be stored in those clusters.

But there are some disadvantages. It requires knowing the size of a file at the time of creation. The file size may be known in some cases like file copying. But, in most cases we cannot predict the file size initially. Also, the disk may contain unusable free space, as they are scattered at different places. Here the file cannot be grown. CD file systems follow the contiguous allocation



Contiguous Allocation of Disk Space



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The figure shows the example of contiguous cluster allocation of files. Here all the four files in the directory has only the starting address and the length. The cluster allocation of each file is contiguous.



Linked Allocation



Each file is a linked list of disk blocks:
blocks may be scattered anywhere on the disk.

Simple – need only starting address

No waste of space

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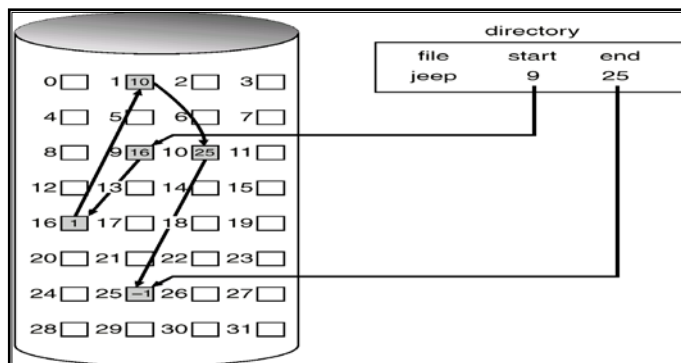
16

Linked allocation can use the available free space in a more effective manner. Here, instead of contiguous allocation, any free block anywhere in the disk can be used.

The directory entry will contain only the starting block address. From the block address stored in the starting block we can go to the next block. Each block allocated will contain the address of the next block. No space is wasted in this allocation.



Linked Allocation



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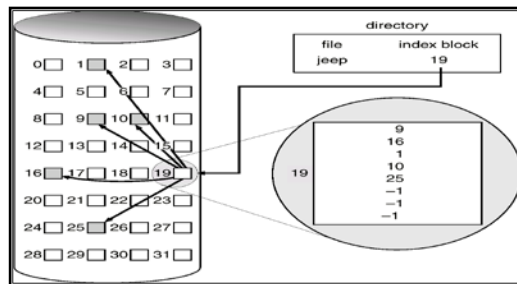
The figure shows an example of Linked cluster allocation of files. In this example the file jeep has only the starting address. When we go the starting address we get the next address and goes on. Finally it ends with a end of file information. In this figure the End of file information is represented with -1.



Indexed Allocation

सी डेक
CDAC

Brings all pointers together into the *index block*.



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In case of the indexed allocation method, a separate index block is maintained to have all the addresses of a file/folder. Each file will have an index block address. The index block contains the addresses of the clusters allocated to the file.

Example shows a file named jeep has index block address of 19. when we look in to the data of the index block, it will contain set of cluster addresses of the file.

Thus indexed allocation helps in direct access of data and prevents external fragmentation as any free block can be used.



FAT File System

सी डेक
CDAC

- FAT - File Allocation Table
- There has been a historic progression of FAT systems from early DOS versions using FAT12 through FAT 16 of the mature DOS and early Windows systems to the final version of FAT, FAT32.
- In each case, the number is the number of bits used to identify a cluster.

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FAT File System is very simple and reliable File system. FAT Stands for File Allocation Table. This file system uses a File Allocation Table where the operating system records how disk space is used. That is, it stores the information about each clusters on the disk.

The FAT file system is relatively uncomplicated, and is supported by virtually all existing operating systems for personal computers

The most common implementations have a serious drawback in that when files are deleted and new files written to the media, directory fragments tend to become scattered over the entire media, making reading and writing a slow process. Defragmentation is one solution to this, but is often a lengthy process in itself and has to be performed regularly to keep the FAT file system clean.



Types of FAT



FAT12

Max. capacity = 2 MB
Mainly for floppy disk

FAT16

Max. capacity = 2 GB
Mainly for DOS

FAT32

Max. capacity = 2 TB
Mainly for Win 9X

There are three different types of FAT based on the partition size. FAT12: Uses 12 bit binary number to represent clusters. A partition formatted using this file system can hold a maximum of 4086 clusters. Floppy disk contains this file system

FAT16: Uses 16 bit binary number to represent clusters. A partition formatted using this file system can hold a maximum of 65526 clusters. The partition with 2GB or less has this file system.

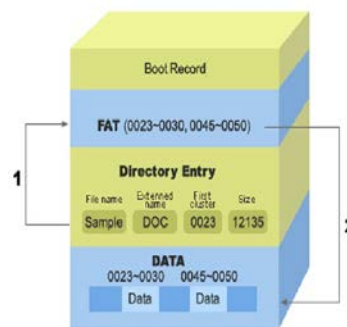
FAT32: This type is supported by newer versions of windows including Windows 2000. It can support drive up to 2TB size. This uses 32 bits to represent clusters.



FAT File System



- Volume boot record
- File allocation tables
- Root directory
- User data area



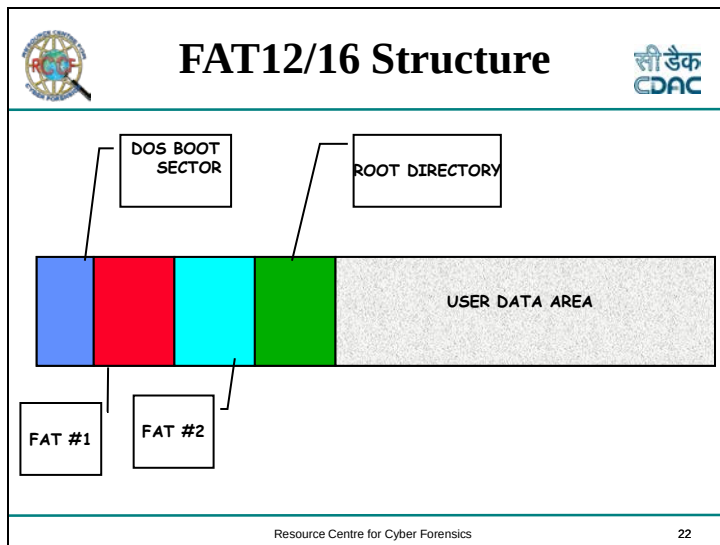
A FAT file system volume is composed of the following four basic regions,

Volume Boot Record: This is the first sector of the volume. This sector contains information like the sector size, no. of sectors per cluster etc.

FAT: Two copies of this table are maintained. This table contains one entry per cluster. This entry uses 12, 16 or 28 bits for FAT12, FAT16 and FAT32.

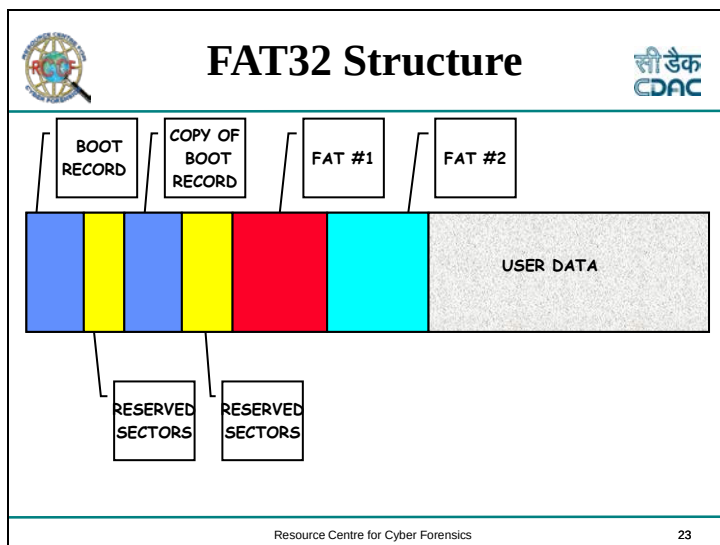
Root Directory: The root directory is kept at a specific location in the volume. This directory contains an entry for each file and folder on the root.

User Data Area: The remaining area on the volume is used to save user files.



The figure shows the Structure of FAT12/16. Both the file system has the same structure.

Here FAT#1 and FAT#2 represents the two copies of FAT. FAT#2 is the duplicate copy of FAT#1.



The figures shows the structure of FAT32. Here the Reserved sectors are the sectors reserved for enhancements of Boot sector. FAT32 has a Copy of Boot record also. No root directory area in FAT32. Root directory area comes in the user data area itself.



Boot Sector



Very first sector on a partition

Has two important functions

- Load in your OS

- Gives all kind of useful information

 - Bytes per sector

 - Sectors per cluster

 - Number of FATs

 - Total number of sectors

 - Max number of root directory entries (FAT12/FAT16)

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The first sector of the FAT partition is Boot Sector. The Boot sector contains Boot load information and the information that the file system uses to access the volume.

Sector Size: The bytes per sector

Cluster Size: The number of sectors per cluster

Total number of FATs : The number of copies of the file allocation table on the volume. Typically this value is 2.

Total number of sectors: Gives the total sectors of the partition.

Total number of root directory entries: The total number of file name entries that can be stored in the root directory of the volume. This field has value only FAT12/16. In case FAT32, the value is zero, because FAT32 has no root directory area.



Boot Sector Cont..



Volume label

Root Directory starting cluster (FAT32)

Copy of Boot Sector Number (FAT32)

File System ID

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Volume Label: A field in the boot sector gives the volume label. Now, the volume label is also stored as special file in the root directory.

Starting cluster of the Root Directory: This field is only defined in FAT32. This is set to the cluster number of the first cluster of the root directory. Data part of Cluster mentioned in the root directory starting cluster number in the Boot sector is Root directory

Copy of Boot Sector Number: If non-zero this field indicates the sector number of the copy of the boot sector. This is defined only in FAT32 file system.

File System ID: Stores the type of the file system.



File Allocation Tables (FAT)



- A FAT Table basically is an array that keeps track of what clusters belong to which file.
- Used to track of the usage of a volume's clusters.
- Each entry for a cluster represents whether it is used, which cluster is next if used, or whether it contains a bad sector.
- Two identical copies of FAT for recovery.

Next part of the FAT file system is FAT. The File Allocation Table maintains a linked list of clusters belonging to a file. It is basically an array that keeps track of what clusters belong to which file.

Also it is used to track of the usage of a volume's clusters. Each entry stores any of the following information to indicate the status of each cluster on the volume

Unused (0x0000)

Cluster in use by a file

Bad cluster (0xFF7)

Last cluster in a file (0xFFF8-0xFFFF)

FAT is followed by one more copy of the entire FAT area for recovery purpose



FAT File Size....



FAT type	Max Clusters	Cluster sizes	Max volume size
FAT12	4,086	0.5 to 4KB	16,736,256 bytes (16MB)
FAT16	65,526	2KB to 32KB	2,147,483,648 bytes (2GB)
FAT32	268,435,456	4KB to 32KB	8,796,093,022,208 bytes (8TB)

 FAT Relationship 		
FAT TYPE	PARTITION SIZE	CLUSTER SIZE (IN BYTES)
FAT12	16 MB	4096
FAT16	32 MB	2048
FAT16	128 MB	2048
FAT16	256 MB	4096
FAT16	512 MB	8192
FAT16	1 GB	16384
FAT16	2 GB	32768
FAT32	<260 MB	512
FAT32	8 GB	4096
FAT32	16 GB	8192
FAT32	32 GB	16384
FAT32	>32 GB	32768

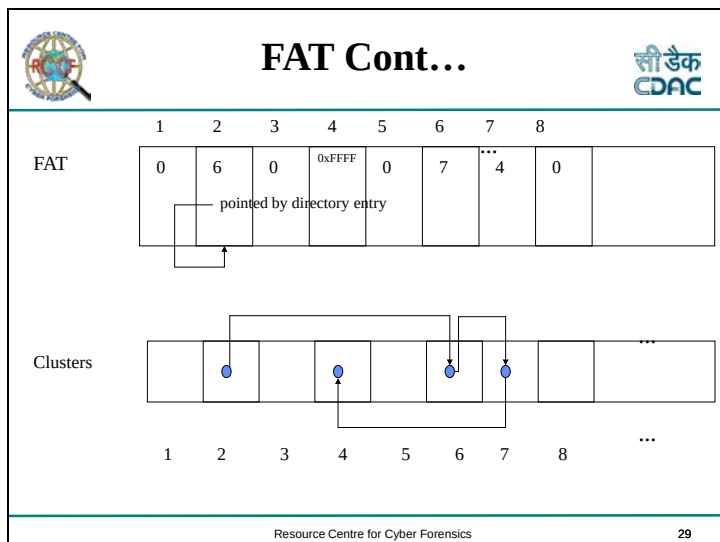
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 FAT Table Entry 			
FAT 12	FAT 16	FAT 32	Meaning
0x000	0x0000	0x0000000	Available
0xFF8-FFF	0xFFF8-FFFF	0xFFFFFFFF8-FFFFFFFF	End of File
0xFF7	0xFFF7	0xFFFFFFFF7	Bad cluster
0xHHH	0xHHHH	0xHHHHHHHH	Next cluster used by file

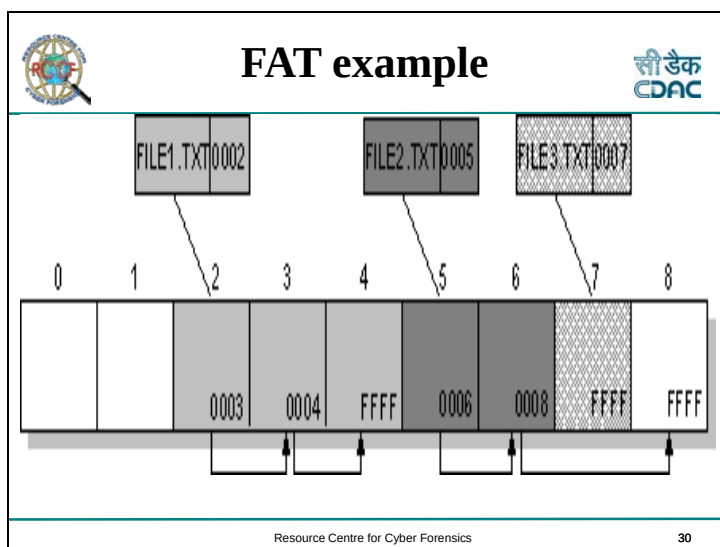
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Table Shows the values in the FAT. There can four types of values. Hexadecimal value 0x0000 represents the corresponding Cluster is free. The Hexadecimal value from 0xFFF8 to 0xFFFF, represents the End of File/Folder.

The Hexadecimal value 0xFFF7 represents Bad cluster. So when we try to create a file this type of cluster will not be allocated. Any other value represents the next cluster number used by the file.



The figure gives an example of FAT for a file/folder having the starting cluster 2. When we read the area of cluster number 2 in FAT we get cluster 6. This means it is the next cluster of the file. When we go to cluster 6 in the FAT area we get cluster number 7 and in the area of Cluster 7 we get 0xFFFF, means End of File. So the Link goes like 2-->6-->7-->4-->EOF



The above figure gives an example of 3 different category of cluster chain

File1.txt – Contiguous Cluster chain –Cluster 2->3->4

File2.txt – Fragmented cluster chain- Cluster 5->6->8

File3.txt – Only one Cluster – Cluster 7



Directory Entry



Keeps track of information about a file or directory.

Each entries are 32Bytes long.

- Short name (name and Extension) entry
- Long name
 - A filename can have 254 characters
- File attributes
- Time Stamps
- File size
- Starting cluster number

Subdirectories are files of same format

After the FAT area Root directory entry comes. This area consists of 32 byte directory entries. Each directory entry will contain the following information:

Name – This field contains a short file name with eight characters for file name and three characters for extension

Long File Name –In case of more than 8 characters of file/folder name, this field will be generated. This contains maximum of 254 characters in unicode form.

File Attributes like system, read only, hidden etc .

Time Stamps like Created Date & time, Modified date & time, Last access date

Starting cluster number of the file/ folder and the **File Size**. The starting cluster number is the address of the first cluster used by the file. Each cluster contains a pointer to the next cluster in the FAT, or an indication (0xFFFF) that this cluster is the end of the file.

Subdirectories also has the same format like files



Directory Entry cont..



Maximum root entries supported

FAT12	224
FAT16	512
FAT 32	Depends on partition size

The table gives the information about maximum root directory entries supported by the FAT File Systems. In case of FAT 12 only 224 directory entries are allowed and in FAT 16 it allows only 512. FAT 32 has no fixed root directory entry area and it can be varied like files and folders.

 Directory Entry cont.. 					
Filename	Ext	Attributes	FileSize	Starting cluster no.	Time Stamps
File1	txt	Read Only, Hidden	400	387	Created Modified Accessed
File2	dll	System	2000	435	Created Modified Accessed
Dir1		Directory	0	005	Created Modified Accessed
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The Table shows an example of a Directory entry with Files and folder

Long File Name

Can be upto 254 characters long
Stored in Directory entry

LFN Part 4
LFN Part 3
LFN Part 2
LFN Part 1
SFN

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The standard directory entry can only support short file names, that is 8 characters for file name and 3 characters for extension. Files with longer names require special directory entries known as Long File Name (LFN) directory entries.

A file will have a normal entry in addition to any LFN entries. Long names are limited to 254 characters. The LFN entries for a file are listed before the short name entry and are in reverse order. One LFN entry can hold 13 unicode Characters. The table gives an example of Long file name divided as four entries ends with Short file name entry



Sub Directories



Same as file entries

Contain two special entries

- .. Has name “..” and refers to parent directory

- . Has name “.” and refers to itself.

Sub directories are also considered as a file entries. Sub directories have two special entries. The first 32 byte entry has a name “.” (dot entry) and the second has a name “..” (dot dot entry).

The dot entry is a directory that points to itself. The dot dot entry is a directory that points to the starting cluster of the parent of this directory.



User Data Area



The rest of the *Partition* contains the most important part, the data area.

The data area sectors are gathered in *clusters*.

All the files and sub directories (beyond the root directory) are stored.

FAT 32 Root directory entries are stored.

The rest of the Partition contains the most important part, the data area. The data area sectors are grouped in to clusters. At least one cluster will be used for a file/folder. All the files and sub directories (beyond the root directory) are stored. Incase of FAT 32, Root directory entries are stored in data area itself.



Write Process



Find a suitable vacant cluster.

Create the directory entry and store the found cluster number as the first cluster.

Find subsequent vacant clusters (if needed).

Update the FAT.

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When an application needs to create a file, it check for vacant clusters. This can be identified from the FAT.

Once the vacant cluster is identified, a directory will be created in the directory entry area with the identified vacant cluster as the starting cluster..

If needed subsequent free clusters can be identified and update the FAT.



Write Process Example



FAT16 partition
Sector/Cluster = 2
Sector size = 512 bytes
Filename – ARMY.TXT (read only)
Size - 1040Bytes
Free clusters – 200 – 400

Step 1

Directory entry

Filename	Ext	Attributes	FileSize	Starting cluster no.	Time Stamps
ARMY	TXT	Read Only	1040	200	16/03/05 16/03/05 16/03/05

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Consider an example of FAT16 partition having

Sectors/Cluster = 2

Sector size = 512 bytes

Free clusters are available from 200 to 400

File to be created is 'Army.txt' (read only) of size 1040Bytes

Step 1 - Creates a directory entry with short filename ARMY.TXT and fills up attributes, file size, starting cluster number and time stamps accordingly.



Write Process Example

(Contd...)



Step 2

	199	200	201	202	203	204	205	206	
FAT	0	201	FFFF	0	0	0	0	0	

Step 3

					200	201					
Data Area											

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Step 2 – Updates the FAT area 200 and 201 . The above example shows that 200 is updated with 201 and 201 with End of file information

Step 3 - Next it fills up the data in the cluster data area 200 and 201. the above example shows the cluster 200 is filled completely and 201 is partially filled.



Read Process



- System information, like file name, size and first cluster number is determined from the directory entry.
- Further cluster numbers are found from the file allocation table (FAT) and read from the corresponding clusters in the data area.

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Here we are going to see, how reading is happening in the system. First it reads the information like filename, size and starting cluster number from the directory entry.

Complete cluster list of the corresponding file will be taken from FAT and read the data from the corresponding clusters in the data area



What happens when a file/folder is deleted?



- To save time on unneeded operations, when a file is deleted, the file content is not deleted.
- In Directory Entry, replace the First character of the short filename with one special character.
- The real change is that the file allocation table marks the clusters used by the file as 'unallocated'.
- If no new data is stored to those clusters, the data can sit there intact for years.

when a file is deleted, to save time on unneeded operations the file content is not deleted. First character of the short file name in the directory entry is replaced by a special character and the FAT entries for the clusters are set to 0 to indicate now those clusters are un allocated. **The file contents are not deleted.** So if that clusters are not allocated again to any new files, the original data can be recovered.



Delete File Example



FAT16 partition
Sector/Cluster = 2
Sector size = 512 bytes
Filename - Army.txt (read only)
Size - 1040Bytes
Free clusters - 200 - 400

Step 1

Directory entry

Filename	Ext	Attributes	FileSize	Starting cluster no.	Time Stamps
~RMY	TXT	Read Only	1040	200	16/03/05 16/03/05 16/03/05

Let us consider that we are deleting the previous example, ARMY.TXT. First it replace the first character of the filename with a special character. May notice from the picture ARMY has changed to ~RMY. No other changes has made in the directory entry.



Delete File Example (Contd...)



Step 2

	199	200	201	202	203	204	205	206
FAT	0	0	0	0	0	0	0	0

Step 3

					200	201				
Data Area										

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Actual changes has happened in the FAT area. It clears all the information corresponding to the file. It fills the Cluster area 200 & 201 in FAT with Zeroes. When we see the data area nothing has changed. Data remains as it is in the data area.



How to recover a deleted file?



- Go to directory entry
- Get the starting cluster no and size of the file
- Calculate no. of clusters occupied by the file
- Go to the data area of that clusters
- Read the data.

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Let us see how we can recover a file / folder. The starting cluster number of the file is available in the directory entry even after deleting the file. So read the starting cluster number and size of the file from the directory entry. Since the FAT area is cleared, the cluster chain for the corresponding file has lost. So calculate the total number of clusters occupied the file by dividing the file size by cluster size. Go to the starting cluster location in the data area and read the data continuously till the size of the file reached..